

Bisect To divide a line into two equal segments. Bisecting a line finds its midpoint.

Equilateral Meaning “equal sides.” We use this term to describe a shape in all which all sides have equal length.

Perpendicular The way we describe two lines that form a 90-degree angle.

Vertex The point at which two edges of a shape meet.

Advanced Graduate-Level Triangle Studies

The scientist and philosopher Gottfried Wilhelm Leibniz thought that our world was the best of all possible worlds because it was the “simplest in hypotheses and richest in phenomena.” He thought that the laws of science could be boiled down to a few simple rules but that those rules led to the complex variety and beauty of the world we observe. This may not be true for the universe, but it is certainly true for triangles. Starting with something that is extremely simple in hypothesis (the idea of a shape with three sides), we enter a world that is extremely rich in phenomena.

Finding the Circumcenter

To begin to see the richness of the phenomena of the world of triangles, consider the following simple algorithm, which you can try with any triangle:

1. Find the midpoint of each side of the triangle.
2. Draw a line from each vertex of the triangle to the midpoint of the vertex’s opposite side.

After you follow this algorithm, you will see something like [Figure 7-5](#).